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# Assignment - 1

## Task 1 - Mathematical Formulation of Formula languages and automata.

1. Input Alphabet, Strings and Formal language

step 1: Identify the input alphabet

The access code contains only two binary Symbols:

$$\Sigma = \{0, 1\}$$

Here:

* 0 - Binary Symbol zero
* 1 - Binary symbol one

step 2: Identify possible strings

Strings are formed by Combining 0's & 1's.

Ex:-

* 0
* 1
* 0 1
* 1 0
* 1 0 1
* 1 1 0 0
* 1 0 1 0 1

step 3: Define the formal language

For this mathematical model, Consider the language of non-Empty binary strings that start & End With the Same Symbol.

$L = \{ w \in \{0,1\}^* \mid w$ starts and Ends with the Same Sy

Examples :-

* 0 - Accepted
* 1 - Accepted
* 101 - Accepted
* 1001 - Accepted
* 011 - Rejected

2. Difference Between Finite strings, Empty strings & languages

| Type | Meaning | Example |
|---|---|---|
| Finite string | A Sequence of Symbol | 1010 |
| Empty string | A string Containing no Symbols | $\varepsilon$ |
| Language | A set of strings | |

1. Finite string :- Contains a limited number of Symbols
2. Empty string :- Contains Zero Symbols. Its length is zero
3. Language :- A collection of Valid string over an alphabet

3. Five - Tuple Representation of Finite Automata

A Finite Automation is represented by five elements :

$$M = (Q, \Sigma, \delta, q_0, F)$$

Sample Binary strings & membership

| Binary string | starts | ends | Result |
|---|---|---|---|
| 0 | 0 | 0 | Accepted |
| 1 | 1 | 1 | Accepted |
| 1 0 1 | 1 | 1 | Accepted |
| 1 0 0 1 | 1 | 1 | Accepted |
| 0 1 1 | 0 | 1 | Rejected |
| 1 0 10 | 1 | 0 | Rejected |

Result:-

The mathematical foundation of the access-code system is defined using binary alphabet, strings, formal language & finite automata

Task 2:- Mathematical Design and construction of DFA

Final states :-

$$F = \{q_{00}, q_{11}\}$$

Five-tuple representation:-

$$M = (Q, \Sigma, \delta, q_0, F)$$

Where:

* $Q$ = set of states.
* $\Sigma$ = Input alphabet.
* $\delta$ = Transition function
* $q_0$ = Initial state
* $F$ = set of final states.

For the access-code System:

$Q = 0$

$F = 1$

$q_0 = q_s$

$\delta$ = Transition function shown in Task 2.

4. Role of states & Transitions.

1. States :- Represent the current condition of the automaton
2. Transitions :- show movement from one state to another based on input.
3. Initial states :- The state where processing begins.
4. Final states :- Indicates that the input string is accepted.

Final state = $q_{11}$ (Final states)

* Input = 100

$q_s \rightarrow q_{11} \rightarrow q_{10} \rightarrow q_{10}$

Final state = $q_{10}$ (Non-final state) ✗

* Input = 0110

$q_s \rightarrow q_{00} \rightarrow q_{01} \rightarrow q_{00} \rightarrow q_{00}$

Final state = $q_{00}$ (Final state) ✓

* Input = 010

$q_s \rightarrow q_{00} \rightarrow q_{01} \rightarrow q_{00}$

Final state = $q_{00}$ (Final state) ✓

*

Result:-

The DFA accepts binary strings that start & End with the Same Symbol & reject strings that start & End with different Symbols.

3. Task-3 NFA for pattern Recognition.

To Construct an NFA that accepts binary strings Containing the Substring 101.

$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{0, 1\}$

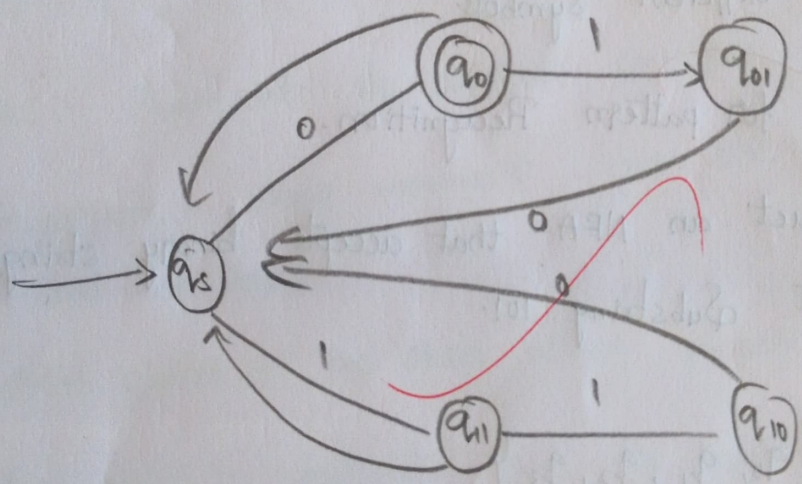
Initial state = $q_0$

Final state = $q_3$

Transition table:-

| State | 0 | 1 |
|---|---|---|
| → $q_s$ | $q_{00}$ | $q_1$ |
| * $q_{00}$ | $q_{00}$ | $q_{01}$ |
| $q_{01}$ | $q_{00}$ | $q_{01}$ |
| $q_{10}$ | $q_{10}$ | $q_{11}$ |
| * $q_{11}$ | $q_{10}$ | $q_{11}$ |

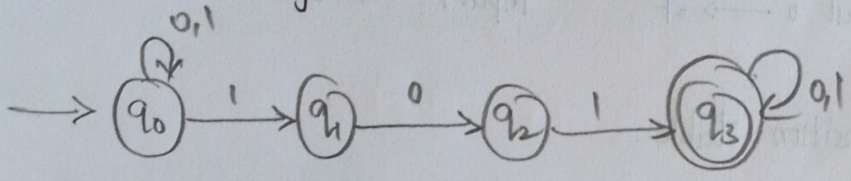
Transition diagram:-




| Input | State |
|---|---|
| start | $q_s$ |
| 1 | $q_{11}$ |
| 0 | $q_{10}$ |
| | $q_{11}$ |

| State | 0 | 1 |
|---|---|---|
| → $q_0$ | $\{q_0\}$ | $\{q_0, q_1\}$ |
| $q_1$ | $\{q_2\}$ | $\emptyset$ |
| $q_2$ | $\emptyset$ | $\{q_3\}$ |
| * $q_3$ | $\{q_3\}$ | $\{q_3\}$ |

Transition diagram:-




## 4. Task-4 Equivalence of NFA and DFA using Subset Construction

$Q = \{q_0, q_1, q_2, q_3\}$

Initial state = $q_0$

Final state = $q_3$

| DFA State | NFA State Subset |
|---|---|
| A | $\{q_0\}$ |
| B | $\{q_0, q_1\}$ |
| C | $\{q_0, q_2\}$ |
| D | $\{q_0, q_1, q_3\}$ |
| E | $\{q_0, q_2, q_3\}$ |
| F | $\{q_0, q_3\}$ |

A = Input 0 → = A  Input 1 → = B

B = Input 0 → = C  Input 1 → = B

C = Input 0 → = A  Input 1 → = D

D = Input 0 → = E  Input 1 → = D

E = Input 0 → = F  Input 1 → = D

F = Input 0 → = F  Input 1 → = D

DFA Transition table :-

| DFA state | 0 | 1 |
|---|---|---|
| → A | A | B |
| B | C | B |
| C | A | D |
| * D | E | D |
| * E | F | D |
| * F | F | D |

Final states :

$F' = \{D, E, F\}$

Compare NFA & DFA outputs

| Input | NFA | DFA |
|---|---|---|
| 101 | Accepted | Accepted |
| 1101 | Accepted | Accepted |
| 100 | Rejected | Rejected |
| 111 | Rejected | Rejected |

5. Task-5 ε-NFA Construction, ε-closure & Equivalence Analysis

$q_0$ → initial state

$q_1, q_2, q_3$ → Indermediate states

$q_4$ → Final states

ε-NFA Transition table :-

| State | 0 | 1 | ε |
|---|---|---|---|
| → $q_0$ | $\{q_0\}$ | $\{q_0\}$ | $\{q_1\}$ |
| $q_1$ | $\emptyset$ | $\{q_2\}$ | $\emptyset$ |
| $q_2$ | $\emptyset$ | $\{q_3\}$ | $\emptyset$ |
| $q_3$ | $\{q_4\}$ | $\emptyset$ | $\emptyset$ |
| * $q_4$ | $\{q_4\}$ | $\{q_4\}$ | $\emptyset$ |

ε-closure :-

| State | ε |
|---|---|
| $q_0$ | $\{q_0, q_1\}$ |
| $q_1$ | $\{q_1\}$ |
| $q_2$ | $\{q_2\}$ |
| $q_3$ | $\{q_3\}$ |
| $q_4$ | $\{q_4\}$ |

NFA Transition table After removing ε

| state | 0 | 1 |
|---|---|---|
| $\rightarrow q_0$ | $\{q_0, q_1\}$ | $\{q_0, q_1, q_2\}$ |
| $q_1$ | $\emptyset$ | $\{q_2\}$ |
| $q_2$ | $\emptyset$ | $\{q_3\}$ |
| $q_3$ | $\{q_4\}$ | $\emptyset$ |
| $*q_4$ | $\{q_4\}$ | $\{q_4\}$ |

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Verification of Equivalence :-

| Input | Result | Status |
|---|---|---|
| 110 | Accepted | Contains 110 |
| 0110 | Accepted | Contains 110 |
| 101 | Rejected | Does not Contain 110 |
| 111 | Rejected | Does not contain 110 |